DO NOW — RECALL

Data Structures · Constructors and Encapsulation

Name: _____ Class: _____ Date: _____

1. How do you create an object in C#?

2. What is an instance field?

3. What does modularity improve?

4. What is LIFO?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Use new with a constructor

2. Model answer

Value held separately per object

3. Model answer

Reuse and maintainability

4. Model answer

Last in first out

By the end, you can…

• Initialise objects with constructors

• Explain private fields and public methods

• Implement basic validated access

Vocabulary — know these terms

Recall the term — constructor

Recall the term — encapsulation

Recall the term — private

Recall the term — public

Recall the term — validation

NEW LEARNING

Read, annotate and explain

Build an object with a valid initial state

```
class Account {
private int balance;
public Account(int opening) {
if (opening < 0)
throw new ArgumentOutOfRangeException();
balance = opening;
}
public int GetBalance() => balance;
}
```

Protect the object invariant

The constructor refuses a negative opening balance before storing it. The field is private, so outside code must use approved public methods instead of bypassing validation.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Construct a valid account; control updates

Constructor sets the initial state

new Account(10) invokes Account(int opening) . It checks 10 is non-negative, then stores 10 in the private field balance .

Method validates the proposed change

savings.Deposit(5) passes the positive-amount check and increases balance from 10 to 15. Outside code cannot change the private field directly.

Check valid and invalid cases

GetBalance() returns 15. Deposit(-3) throws an exception before changing the balance. Encapsulation is controlled access, not merely writing private .

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Trace a different validated account

Using the Account class, run Account current = new Account(20); current.Deposit(7); Then attempt current.Deposit(-2); . What does current.GetBalance() give afterwards if the exception is caught by the caller?

Method / working:

2. Construct a different valid object

A Box has private int width; and a constructor accepting width. Write the validation condition that rejects zero or negative widths, then state which values new Box(4) and new Box(0) should allow.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Write a constructor for class Box that stores integer width. [4]

Question 2

Apply independently
Explain encapsulation. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Suggest validation for a bank deposit method. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: a constructor has a declared return type void.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.